Yug Dhawan

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New Delhi, India (IST, overlaps EU hours and US mornings) | Backend | AI/LLM | ML

Summary

Backend and AI engineer finishing a B.Tech (2027). During an SDE internship I shipped a production CRM, with an LLM advisor and Meta Ads lead intake, that 50+ agents and managers use every day. I build RAG and LLM pipelines, multimodal ML models and high-throughput backends in Python, Go and C++. Top 10% of 7,000+ teams in the Amazon ML Challenge 2025. A Discord moderation bot I built in school protected 320,000 users across 150+ servers.

Technical Skills

Languages: Python, Go, C++, JavaScript, TypeScript, Dart, SQL
AI/ML: RAG, LLM Integration (Claude, OpenAI, Groq), Vector DBs, PyTorch, HuggingFace, CLIP, DINOv2, LightGBM, scikit-learn
Backend: REST APIs, FastAPI, Node.js, Supabase/PostgreSQL (RLS), Event-Driven Systems, Concurrent Pipelines
Cloud/DevOps: Docker, AWS, CI/CD, GitHub Actions, Git, Linux
Mobile and Security: Flutter, E2E Encryption (RSA-4096 + AES-256-GCM), OWASP Top 10 Remediation

Experience

Agnayi Infra Resources Jun 2026 – Oct 2026
SDE Intern *New Delhi, India*

- Built a CRM integrating Meta Ads lead scheduling and property listings, adopted by 50+ agents and managers to track performance by visits and revenue generated.
- Added an in-app AI advisor (Groq LLM with Tavily web research and file uploads), so agents get answers without leaving the CRM.
- Building BrokerBro, a social app for brokers, across a Flutter client and a Supabase backend with Postgres row-level security, a 3-tier access model and end-to-end encrypted messaging; remediated 24 findings from an OWASP-aligned security audit.

Projects

Agentic UPI Transaction Categorization Engine 2024 – Present
Go, Python, Claude API, OpenAI API, PostgreSQL

- Classified 50,000+ raw UPI transaction strings into categories with natural-language summaries; a vector database does semantic merchant matching (RAG), and only unresolved strings reach the LLM, on a zero-GPU stack.

UPI City: Fraud Detector Test Bench 2026
Go, Python, PyOD, scikit-learn

- Simulated a payment network with injected fraud to score detectors against ground truth: graded 8 detectors over 123,622 payments (4,463 fraudulent); 4 scored below chance, and my fused detector reached 15.1x lift over chance.

Amazon ML Challenge 2025: Multimodal Pipeline 2025
PyTorch, HuggingFace, CLIP, DINOv2, LightGBM

- Top 10% of 7,000+ teams with two multimodal pipelines built in about 30 hours; a 50-thread fetcher ingested 75,000 images; DINOv2 + MiniLM and CLIP + LightGBM regression models, cross-validated.

Private Photo Search 2026
Python, CLIP, HDBSCAN, OpenCV

- Natural-language search over 4,958 photos in 13 to 17 ms per query, unsupervised albums, and face grouping (546 faces clustered in under 0.1 s), all running locally.

High-Performance Multithreaded Web Server 2024 – Present
C++17, epoll, eventfd, GoogleTest

- Scaled an HTTP/1.1 server from one epoll loop to one per core with EPOLLEXCLUSIVE: 158k to 665k requests/sec at 1,000 keep-alive connections (5.5 ms p99, 11 MB RSS), on par with 28-worker nginx on the same machine.

Education

Bachelor of Technology (B.Tech) in Electronics and Communication Engineering 2023 – 2027
Maharaja Agrasen Institute of Technology (GGSIPU), New Delhi

Achievements & Publications

Top 10% Finisher, Amazon ML Challenge 2025 (7,000+ teams)
Protected (Discord moderation bot, 2022): 320,000 users across 150+ servers
Sharma et al., IJSREM (2025): AI-Driven Workforce Transformation